SENMARCK ENERGY

Das Energiespeichersystem der Zukunft - senmarck.ch

Senmarck Residential Storage Solution Guard Plus Series

Powered by CATL Battery

12 Gründe für das Energiespeicher System von Senmarck:

Hohe Leistung:
Der Wechselrichter bietet eine Leistung von 20 kWp Leistung.

Sicherheit:
SELV (Safety Extra Low Voltage) und PELV (Protection Extra Low Voltage) garantieren Sicherheit für Spannungen unter 60V.

Langlebige Batterie:
CATL-Batterie mit 10'000 Batteriezyklen.

Vielseitigkeit:
Geeignet für die Ausseninstal-lation und Heizung für extreme Temperaturen.

Inselfähigkeit und Kaltstart:
Betrieb im Inselbetrieb und und mit Kaltstart möglich.

Ausbaufähig:
Pro Wechselrichter erweiterbar auf 40 kWh Speicherkapazität.

USV/USV-Funktion:
Völlig unabhängiges USV-System ohne FRT-Verpflichtung in Österreich.

Eingebaute Transfer Schaltkasten:
Integrierte Umschaltbox vermeidet Kosten und Überlastungen.

Benutzerfreundliche App:
Fernsteuerung, historische Daten, und umfassende Statistiken.

Web-Plattform:
Zusätzliche Option für Daten Überwachung und Steuerung.

Kompatible Schnittstelle:
Kompatibel mit SolarManPV Datenloggern und LAN-Option anstelle von WLAN.

Support:
In der Schweiz.

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Hybrid Inverter Model | Guard Plus 10KH3

PV String Input

Max. Continuous PV Input Power	20kW
Max. DC Voltage	1100V
Nominal Voltage	720V
MPPT Voltage Range	140V-1000V
MPPT Voltage Range (Full Load)	420V-850V
Start Voltage[1]	200V
Number of MPPT	2
Strings Per MPPT	1
Max. Input Current Per MPPT	15A
Max. Short-circuit Current Per MPPT	20A

Output (On Grid)

Nominal AC Output Power	10kW
Max. AC Apparent Power	11kVA
Nominal AC Voltage	400Vac
AC Grid Frequency Range	50 / 60Hz±5Hz
Nominal Output Current	14.5A
Max. Output Current	16A
Power Factor (cosΦ)	0.8leading-0.8lagging *
Max. total harmonic distortion	＜3%

Battery Input

Battery Type	LFP (LiFePO4)
Nominal Battery Voltage	51.2V
Charging Voltage Range	44-58V
Max. Charging Current	160A
Max. Discharging Current	200A
Battery Capacity	200/400/600/800Ah

Output (Off Grid)

Nominal AC Output Power	9.2kW
Max. Apparent Power	10kVA
Nominal Output Current	13.3A
Max. Output Current	14.5A
Nominal Output Voltage	400V
Nominal Output Frequency	50/60Hz
Output THDv (@Linear Load)	<3% (Linear Load)
Power factor range	0.8leading-0.8lagging

Efficiency

Max. PV Efficiency	97.60%
Euro. PV Efficiency	97.00%

Features & Protections

Anti-islanding Protection	Yes
Output Over Current Protection	Yes
DC Reverse Polarity Protection	Yes
String Fault Detection	Yes
AC/DC Surge Protection	DC Type II; AC Type III
Insulation Detection	Yes
AC Short Circuit Protection	Yes
Input-side disconnection device	Yes
Residual current monitoring	Yes
AC overcurrent protection	Yes
AC overvoltage protection	Yes
Ripple receiver control	Yes
Battery reverse charging from grid	Yes

General Data

Dimensions W x H x D	540*980*240mm
Weight	49kg
Operating Temperature Range	-25℃~+60℃
Cooling Type	Natural Convection
Max. Operation Altitude	2000m
Operation Humidity	0~95% (No Condensation)
IP Class	IP65
Topology	Battery Isolation
Communication	RS485/CAN2.0/WIFI/4G
Display	LCD/APP

Standard Compliance (more available upon request)

Certificates	IEC 62109-1: 2010; EN IEC 61000-6-1: 2019 EN IEC 61000-6-2: 2019; EN IEC 61000-6-4: 2019 EN 61000-6-3:2007/A1: 2011; EN IEC 61000-3-11:2019 IEC 62109-2:2011
Grid connection standards	VDE-AR-N 4105:2018 DIN VDE V 0124-100:2020; CEI 0-21:2019; OVE-Richtlinie R 25:2020; EN 50549-1:2019/AC:2019

* 0.95leading-0.95lagging for Germany.

Battery Model | Guard Plus-Pack 5.1

Performance

Energy Capacity	5.12kwh
Usable Capacity	4.6kwh
Depth of Discharge (DoD)	90%
Nominal Voltage	51.2V
DC Circuit Breaker	125A
Operating Voltage Range	44.8-56.5V
Internal Resistance	<20mΩ
Cycle Life	10,000cycle
Battery type	LiFePO4
System Weight	54KG
Dimension (W x D x H)	540*490*240
IP Protection	IP65
Warranty	5 Years Product Warranty, 10 Years Performance Warranty

General Specification

Max. Charge/Discharge Current	50A/80A
Rated DC power	4096W
Max. Charge/Discharge Power	2825W/4096W
Operating Temperature Range	0 to 50℃ charging -10 to 50℃ discharging

General Specification

Humidity	0~95% (No condensation)
Capacity	200-800Ah
Power Consumption	<2W
Communication	CAN & RS485
Monitoring Parameters	System voltage, current, cell voltage, cell temperature, PCBA temperature measurement
Cooling	Natural convection
Noise emission	≤25db
Max. operating altitude	≤2000m
Display	SOC status indicator
Installation	Floor stand, Wall mount(optional)
Scalability	Max. 4 battery packs in parallel operation for single phase system; Max. 8 battery packs in parallel operation for 3 phase system;
Compatibe inverters	Guard Plus 3680E/5000E, Guard Plus 8KH3/10KH3/12KH3

Standard Compliance (more available upon request)

Certificates	CE, EMC, IEC62619, UN38.3, UL1973

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